T3 matched EFT Lagrangian terms: EFT_1_after_F

EFT order: 5; loop order: 1. Rows are model configurations and columns are field configurations. This stage integrates out F. Remaining active T3 fields: S1, S2.

Notation key

symbol	interaction / definition
g_1	$D_\mu \supset i g_1 Y B_\mu$
g_2	$D_\mu \supset i g_2 T^A W_\mu^A$
g_3	$D_\mu \supset i g_3 t^A G_\mu^A$
Y_u	$-\bar{Q} Y_u \widetilde{H} u + \text{h.c.}$
Y_d	$-\bar{Q} Y_d H d + \text{h.c.}$
Y_e	$-\bar{L} Y_e H e + \text{h.c.}$
y_1	$L F S_1 + \text{h.c.}$
y_2	$L F S_2 + \text{h.c.}$
M_F	$-\bar{F} M_F F$
T	$\mathrm{Tr}(Y_e^\dagger Y_e + 3 Y_u^\dagger Y_u + 3 Y_d^\dagger Y_d)$
$T_\nu^{(1)}$	$\mathrm{Tr}(y_1^\dagger y_1)$
$T_\nu^{(2)}$	$\mathrm{Tr}(y_2^\dagger y_2)$
T_ν	$T_\nu^{(1)} + T_\nu^{(2)}$
m_1^2	$m_1^2 S_1^\dagger S_1$
m_2^2	$m_2^2 S_2^\dagger S_2$
λ_1	$\frac{1}{2}\lambda_1 (H^\dagger H)^2$
$\lambda_{S_1}^{(1)}$	$\frac{1}{2}\lambda_{S_1}^{(1)} (S_1^\dagger S_1)^2$
$\lambda_{S_2}^{(1)}$	$\frac{1}{2}\lambda_{S_2}^{(1)} (S_2^\dagger S_2)^2$
$\lambda_{HS_1}^{(1)}$	$\lambda_{HS_1}^{(1)} (H^\dagger H)(S_1^\dagger S_1)$
$\lambda_{HS_2}^{(1)}$	$\lambda_{HS_2}^{(1)} (H^\dagger H)(S_2^\dagger S_2)$
$\lambda_{HS_1}^{(A)}$	$\lambda_{HS_1}^{(A)} (H^\dagger T^A H)(S_1^\dagger T^A S_1)$
$\lambda_{HS_2}^{(A)}$	$\lambda_{HS_2}^{(A)} (H^\dagger T^A H)(S_2^\dagger T^A S_2)$
λ_5	$\lambda_5 H H S_1 S_2^\dagger + \text{h.c.}$
$\lambda_{12}^{(1)}$	$\lambda_{12}^{(1)} (S_1^\dagger S_1)(S_2^\dagger S_2)$
$\lambda_{12}^{(A)}$	$\lambda_{12}^{(A)} (S_1^\dagger T^A S_1)(S_2^\dagger T^A S_2)$
$\lambda_{12}^{(\times)}$	$\lambda_{12}^{(\times)} (S_1^\dagger S_2)(S_2^\dagger S_1)$
$\lambda_{S_1}^{(A)}$	$\lambda_{S_1}^{(A)} (S_1^\dagger T^A S_1)(S_1^\dagger T^A S_1)$
$\lambda_{S_2}^{(A)}$	$\lambda_{S_2}^{(A)} (S_2^\dagger T^A S_2)(S_2^\dagger T^A S_2)$
α	$Y_{S_1} = \alpha/2,\ Y_F = (\alpha+1)/2,\ Y_{S_2} = (\alpha+2)/2$
$C_2(d)$	$(d^2-1)/4$

$T(d)$	$d(d^2-1)/12$
$C^{ij;AB}_{LLS_1S_2}$	$L_iL_jS_1^AS_2^B$
C_{ijab}	$L_iL_j\phi_a\phi_b$
C_5^{ij}	$(L_i^TCL_j)HH$
R_W	$16\pi^2\beta_{C_5} \supset R_W\lambda_5 C_{LLS_1S_2}$

Field combinations 1 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2}{\epsilon}$ $-C_{S_1^\dagger S_1}S_1S_1{}^\dagger$ $+C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger}{\epsilon}$ $+\frac{1}{2}S_1^2\left(S_1{}^\dagger\right)^2\lambda^{(1)}_{S_1}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda^{(1)}_{S_1}}{\epsilon}$	$+D_{\mu_1}(S_1)^\dagger D_{\mu_1}(S_1)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S^2_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$ $-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}}{\sqrt{3}}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2}{\epsilon}$ $-C_{S_1^\dagger S_1}S_1S_1{}^\dagger$ $+C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger}{\epsilon}$ $+\frac{1}{2}S_1^2\left(S_1{}^\dagger\right)^2\lambda^{(1)}_{S_1}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda^{(1)}_{S_1}}{\epsilon}$	$+D_{\mu_1}(S_1)^\dagger D_{\mu_1}(S_1)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S^2_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$ $-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2}{\epsilon}$ $-C_{S_1^\dagger S_1}S_1S_1{}^\dagger$ $+C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger}{\epsilon}$	$+D_{\mu_1}(S_1)^\dagger D_{\mu_1}(S_1)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$ $-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$
cont.								$+\frac{1}{2}S_1^2\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}}{\epsilon}$		$+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
cont.								$-\hbar\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}$
cont.										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$ $+\frac{1}{2}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{\epsilon}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$ $+\frac{1}{2}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{\epsilon}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$
cont.								$+\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}}{\epsilon}$		$+\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\epsilon}$
cont.								$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$		$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$ $+\frac{1}{2}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\epsilon}$	$+D_{\mu_1}(S_{1,j_1})^\dagger D_{\mu_1}(S_{1,j_1})$	$-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\epsilon}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
								$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{\epsilon}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger}{\epsilon}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger}{\epsilon}$
cont.								$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$ $+\frac{1}{2}\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger$ $+\frac{\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-C_{S_2^\dagger S_2}S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger$ $+\frac{\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
								$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger}{\epsilon}$		$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger}{\epsilon}$
cont.								$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$ $+\frac{1}{2}\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger$ $+\frac{\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-C_{S_2^\dagger S_2}S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger$ $+\frac{\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
cont.								$+\frac{1}{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$+\frac{1}{3}\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
								$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger}{\epsilon}$		$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger}{\epsilon}$
cont.								$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$ $+\frac{1}{2}\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger$ $+\frac{\sqrt{3}C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-C_{S_2^\dagger S_2}S_{2,j_1}{S_{2,j_1}}^\dagger$ $+\frac{1}{2}\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger$ $+\frac{\sqrt{3}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\epsilon}$ $-\frac{5\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
cont.								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
								$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger}{\epsilon}$		$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger}{\epsilon}$
cont.								$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left({S_{1,j_1}}^\dagger\right)^2}{3\epsilon}$ $-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$ $+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$ $+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_2{}^\dagger\right)^2}{\epsilon}$ $-C_{S_2^\dagger S_2}S_2S_2{}^\dagger$ $+C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}S_2{}^\dagger$ $+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}S_2{}^\dagger}{\epsilon}$ $+\frac{1}{2}S_2^2\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2{\lambda_{S_2}^{(1)}}^\dagger}{\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^\dagger$		

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}}$		$-\hbar\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2$		
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}}$		
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2}{3\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_2{}^\dagger\right)^2}{\epsilon}$
								$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$		$-C_{S_2^\dagger S_2}S_2{S_2}^\dagger$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger$
								$+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$		$+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger}{\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$		$+\frac{1}{2}S_2^2\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}}$		$-\hbar\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2$		
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}}$		
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2}{3\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_2{}^\dagger\right)^2}{\epsilon}$
								$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$		$-C_{S_2^\dagger S_2}S_2{S_2}^\dagger$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger$
								$+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$		$+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger}{\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$		$+\frac{1}{2}S_2^2\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}}{\epsilon}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}}$		$-\hbar\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left(S_2{}^\dagger\right)^2\lambda_{S_2}^{(1)}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2$		
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}\epsilon}$		
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}}$		
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2}{3\epsilon}$		$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$
								$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$		$-C_{S_2^\dagger S_2}S_{2,j_1}{S_{2,j_1}}^\dagger$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
								$+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$		$+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$		$-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1}{\sqrt{3}}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2}{\sqrt{3}}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left({S_{1,j_1}}^\dagger\right)^2}{3\epsilon}$		$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_{2,j_1}}^\dagger\right)^2}{3\epsilon}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-C_{S_2^\dagger S_2}S_{2,j_1}{S_{2,j_1}}^\dagger$
								$+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$		$+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$
								$+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$		$+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$		$-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1}{\sqrt{3}}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2}{\sqrt{3}}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left({S_{1,j_1}}^\dagger\right)^2}{3\epsilon}$		$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_{2,j_1}}^\dagger\right)^2}{3\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-C_{S_1^\dagger S_1}S_{1,j_1}{S_{1,j_1}}^\dagger$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$-C_{S_2^\dagger S_2}S_{2,j_1}{S_{2,j_1}}^\dagger$
								$+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$		$+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$
								$+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$		$+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{3\epsilon}$		$-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}^1}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(1)}}^1}{\sqrt{3}}$
								$+\frac{1}{2}S_{1,j_3}S_{1,j_4}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2$		$+\frac{1}{2}S_{2,j_3}S_{2,j_4}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}\epsilon}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(2)}}^2}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{\lambda_{S_2}^{(2)}}^2}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	$S_1\,D$	S_2
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S^2_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_{1,j_1}{}^\dagger\right)^2}{3\epsilon}$ $-C_{S_1^\dagger S_1}S_{1,j_1}S_{1,j_1}{}^\dagger$ $+\frac{C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2C_{S_1^\dagger S_1}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$+D_{\mu_1}\left(S_{1,j_1}\right)^\dagger D_{\mu_1}\left(S_{1,j_1}\right)$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S^2_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2}{3\epsilon}$ $-C_{S_2^\dagger S_2}S_{2,j_1}S_{2,j_1}{}^\dagger$ $+\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$
cont.								$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_1}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_1}}{\sqrt{3}\epsilon}$		$-\frac{4\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_1}}{\sqrt{3}}$ $+\frac{1}{2}S_{1,j_3}S_{1,j_4}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_1}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_1}}{\sqrt{3}\epsilon}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_2}}{\sqrt{3}}$ $+\frac{1}{2}S_{2,j_3}S_{2,j_4}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}}{\sqrt{3}\epsilon}$
cont.								$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_1}}{\sqrt{3}}$		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda^{(2)}_{S_2}}{\sqrt{3}}$

Field combinations 2 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_2 D	$B_{\mu\nu}$	$W_{\mu\nu}$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{9\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{\epsilon}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$		
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{\epsilon}$	
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{3\hbar B^2_{\mu_1\mu_2}}{\epsilon}$	$-\frac{2\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{\epsilon}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$		$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{\epsilon}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{3\hbar B^2_{\mu_1\mu_2}}{\epsilon}$	$-\frac{2\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{\epsilon}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+D_{\mu_1}\left(S_2\right)^\dagger D_{\mu_1}\left(S_2\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_2\right)^\dagger D_{\mu_1}\left(S_2\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+D_{\mu_1}\left(S_2\right)^\dagger D_{\mu_1}\left(S_2\right)$	$-\frac{9\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{9\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+D_{\mu_1}\left(S_{2,j_1}\right)^\dagger D_{\mu_1}\left(S_{2,j_1}\right)$	$-\frac{9\hbar B^2_{\mu_1\mu_2}}{2\epsilon}$	$-\frac{\hbar\left(W^{A_1}_{\mu_1\mu_2}\right)^2}{2\epsilon}$

Field combinations 3 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$S_1\,H$	$S_1\,S_2$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger$ $+H_{a_2}S_1H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+S_1S_{2,j_2}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\sqrt{3}}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger$ $+H_{a_2}S_1H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+S_1S_{2,j_2}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\sqrt{3}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_1{}^\dagger$ $+H_{a_2}S_1H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_1S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_1{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+S_1S_{2,j_2}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}H_{a_1}{}^\dagger S_1{}^\dagger\mathcal{I}_{H1,a_1,a_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\epsilon}$ $-\hbar\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}S_1{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda^{(1)}_{12}}{\sqrt{3}}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$ $+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$ $+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
cont.								$+\frac{2\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$ $+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1 H	S_1 S_2
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}}{\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
											$-\sqrt{3}S_{1,j_2}S_{2,j_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+\sqrt{3}S_{1,j_1}S_{2,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
cont.											$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
cont.											$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
											$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_5\mathcal{C}_{LS_2F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}}{\sqrt{3}\epsilon}$	$+\mathrm{CNewScalar1NewScalar2}\hbar S_{1,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{j_1,j_2}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$+\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}S_{1,j_1}y_{1,i_1,r_1}\lambda_5\mathcal{C}_{LS_2F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}}{\sqrt{3}}$	$-\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}$
								$+\frac{2\hbar\overline{\lambda_5}{L_{a_3,i_1}}C P_R{\overline{L_{a_4,i_2}}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1F^*_{a_2,j_1}}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\epsilon}$	$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}}{\epsilon}$
cont.										$-2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1F^*_{a_2,j_1}}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}\mathrm{Bar(cg1)}_{j_3,j_4}}{\epsilon}$
										$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_2}\mathrm{Bar(cg1)}_{j_1,j_2}\mathrm{Bar(cg1)}_{j_3,j_4}}{\epsilon}$
										$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar(cg1)}_{j_2,j_3}{S_{1,j_1}}^\dagger}{\epsilon}$
cont.										$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_2}\mathrm{Bar(cg1)}_{j_2,j_3}{S_{1,j_1}}^\dagger}{\epsilon}$
								$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$	$+\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_1}\mathrm{Bar(cg1)}_{j_2,j_3}{S_{2,j_1}}^\dagger}{\epsilon}$
								$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$	$-2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_2}\mathrm{Bar(cg1)}_{j_2,j_3}{S_{2,j_1}}^\dagger$
								$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$+\hbar\overline{\mathrm{CNewScalar1NewScalar2}}\mathrm{cg1}_{j_1,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}}{\epsilon}$	$-\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
								$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
											$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
cont.											$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
											$-2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
											$+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.											$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
											$+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	$S_1\,H$	$S_1\,S_2$
											$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg1}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger}{\epsilon}$
cont.											$-2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}\mathrm{cg1}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger$
											$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\mathrm{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{S_{2,j_3}}^\dagger}{\epsilon}$
cont.											$-2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}\mathrm{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{S_{2,j_3}}^\dagger$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg1}_{j_1,j_3}\mathrm{cg1}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger{S_{2,j_4}}^\dagger}{\epsilon}$
											$+\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}\mathrm{cg1}_{j_1,j_3}\mathrm{cg1}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger{S_{2,j_4}}^\dagger$
											$+S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$
cont.											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$
											$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$
											$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg1})_{j_2,j_3}{S_{1,j_1}}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg1})_{j_2,j_3}{S_{1,j_1}}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.											$-\sqrt{3}S_{1,j_2}S_{2,j_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+\sqrt{3}S_{1,j_1}S_{2,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.											$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
cont.											$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+\frac{2\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
cont.											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg1})_{j_2,j_3}S_{2,j_1}{}^\dagger\lambda_{S_2}^{(1)}}{\epsilon}$
cont.											$+\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg1})_{j_2,j_3}S_{2,j_1}{}^\dagger\lambda_{S_2}^{(1)}$
											$+\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{S_{2,j_3}}^\dagger\lambda_{S_2}^{(1)}}{\epsilon}$
cont.											$+\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{S_{2,j_3}}^\dagger\lambda_{S_2}^{(1)}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\epsilon}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
								$+\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_1}^{(1)}$	$-2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1 H	S_1 S_2
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$	$+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.										$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$	$+2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$+S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$	
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}}{\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
										$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
											$-\sqrt{3}S_{1,j_2}S_{2,j_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+\sqrt{3}S_{1,j_1}S_{2,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
cont.											$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
cont.											$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
											$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}}{\epsilon}$
cont.											$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda^{(2)}_{12}$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$		$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$		$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{10\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$-\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3\epsilon M_F}$		$+\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}{}^1}{\sqrt{3}\epsilon}$	$-\frac{10}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$+\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}{}^1}{\sqrt{3}}$	$+\frac{2}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.								$-\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$		$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$	$+\frac{8}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}}{\epsilon}$	$+\frac{2}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_1}$	$+S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
cont.										$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
										$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}}{\epsilon}$	$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}}{\epsilon}$
cont.										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_1}$	$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda^{(1)}_{12}$
											$-\sqrt{3}S_{1,j_2}S_{2,j_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
											$+\sqrt{3}S_{1,j_1}S_{2,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$
cont.											$+\frac{3\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}}{\epsilon}$
											$+\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{12}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	$S_1\,H$	$S_1\,S_2$
											$-\frac{3\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.											$-\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+\frac{3\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$+\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
cont.											$-\frac{3\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
cont.											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
cont.											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$-\frac{2\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_5\mathrm{Bar}(\mathrm{cg1})_{a_1,j_1,a_2,j_2}}{\epsilon}$	$+\mathrm{CNewScalar1NewScalar2}\hbar S_{1,j_1}S_{2,j_2}\mathrm{Bar}(\mathrm{cg3})_{j_1,j_2}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$-\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_5\mathrm{Bar}(\mathrm{cg1})_{a_1,j_1,a_2,j_2}$	$-\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg3})_{j_1,j_2}$
								$-\frac{2\hbar\overline{\lambda_5}L_{a_3,i_1}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3\epsilon M_F}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg3})_{j_1,j_2}}{\epsilon}$
										$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{5\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar}(\mathrm{cg3})_{j_1,j_2}\mathrm{Bar}(\mathrm{cg3})_{j_3,j_4}}{9\epsilon}$
cont.								$-\frac{2\hbar\overline{\lambda_5}L_{a_3,i_1}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$	$-\frac{2\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{\epsilon}$	$+\frac{1}{9}\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_2}\mathrm{Bar}(\mathrm{cg3})_{j_1,j_2}\mathrm{Bar}(\mathrm{cg3})_{j_3,j_4}$
										$-\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger$	$+\frac{10\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar}(\mathrm{cg3})_{j_2,j_3}{S_{1,j_1}}^\dagger}{3\sqrt{3}\epsilon}$
										$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}\epsilon}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_2}\mathrm{Bar}(\mathrm{cg3})_{j_2,j_3}{S_{1,j_1}}^\dagger}{3\sqrt{3}}$
cont.										$+\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$	$+\frac{10\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_1}\mathrm{Bar}(\mathrm{cg3})_{j_2,j_3}S_{2,j_1}{}^\dagger}{3\sqrt{3}\epsilon}$
										$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_2}\mathrm{Bar}(\mathrm{cg3})_{j_2,j_3}S_{2,j_1}{}^\dagger}{3\sqrt{3}}$
										$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_1}^{(1)}}{\epsilon}$	$+\hbar\overline{\mathrm{CNewScalar1NewScalar2}}\mathrm{cg3}_{j_1,j_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_1}^{(1)}$	$-\frac{1}{2}C_{S_2^\dagger S_2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
										$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$	$-\frac{C_{S_2^\dagger S_2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{\epsilon}$
											$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
cont.										$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\epsilon}$	$-\frac{10\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$	$-\frac{10}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
											$+\frac{2}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.											$+\frac{8\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
											$-\frac{10\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
											$+\frac{8}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.											$+\frac{2}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
											$+\frac{10\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg3}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger}{3\sqrt{3}\epsilon}$
											$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}\mathrm{cg3}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger}{3\sqrt{3}}$
cont.											$+\frac{10\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\mathrm{cg3}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger}{3\sqrt{3}\epsilon}$
											$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}\mathrm{cg3}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger}{3\sqrt{3}}$
cont.											$-\frac{5\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg3}_{j_1,j_3}\mathrm{cg3}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger S_{2,j_4}{}^\dagger}{9\epsilon}$
											$+\frac{1}{9}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}\mathrm{cg3}_{j_1,j_3}\mathrm{cg3}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger S_{2,j_4}{}^\dagger$
											$+S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1 H	S_1 S_2
											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1}{\epsilon}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1$
cont.											$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}{S_{1,j_1}}^\dagger{\lambda_{12}^{(2)}}}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}{S_{1,j_1}}^\dagger{\lambda_{12}^{(2)}}}{\sqrt{3}}$
											$-\sqrt{3}S_{1,j_2}S_{2,j_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}$
cont.											$+\sqrt{3}S_{1,j_1}S_{2,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}$
											$+\frac{3\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}}{\epsilon}$
											$+\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}$
cont.											$-\frac{3\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}}{\epsilon}$
											$-\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}$
											$+\frac{3\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}}{\epsilon}$
cont.											$+\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}$
											$-\frac{3\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}}{\epsilon}$
											$-\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\lambda_{12}^{(2)}}$
											$+\frac{2\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger{\lambda_{12}^{(2)}}}{\sqrt{3}\epsilon}$
cont.											$+\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger{S_{2,j_3}}^\dagger{\lambda_{12}^{(2)}}}{\sqrt{3}}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2$
											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2}{\epsilon}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2$
cont.											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}{S_{2,j_1}}^\dagger{\lambda_{S_2}^{(1)}}}{\epsilon}$
											$+\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}{S_{2,j_1}}^\dagger{\lambda_{S_2}^{(1)}}$
cont.											$+\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{S_{2,j_3}}^\dagger{\lambda_{S_2}^{(1)}}}{\epsilon}$
											$+\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{S_{2,j_3}}^\dagger{\lambda_{S_2}^{(1)}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3\epsilon M_F}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{10\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
								$-\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{3\epsilon M_F}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}\epsilon}$	$-\frac{10}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
									$+\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$	$+\frac{2}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.								$-\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{3M_F}$	$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1{\lambda_{HS_1}^{(1)}}$	$+\frac{8}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
									$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1{\lambda_{HS_1}^{(1)}}}{\epsilon}$	$+\frac{2}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
									$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1{\lambda_{HS_1}^{(1)}}$	$+S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1$
cont.										$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2{\lambda_{HS_1}^{(2)}}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1}{\epsilon}$
										$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2{\lambda_{HS_1}^{(2)}}}{\epsilon}$	$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1 H	S_1 S_2
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1}{\epsilon}$
cont.										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2$	$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(1)}}^1$
											$-\sqrt{3}S_{1,j_2}S_{2,j_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+\sqrt{3}S_{1,j_1}S_{2,j_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
cont.											$+\frac{3\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$+\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$-\frac{3\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
cont.											$-\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+\frac{3\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$+\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
cont.											$-\frac{3\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}}{\epsilon}$
											$-\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2$
cont.											$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2$
cont.											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2}{\epsilon}$
											$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{\lambda_{12}^{(2)}}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_2}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_2}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$
								$+\frac{2\hbar\overline{\lambda_5}{L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^1}{\epsilon}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_2}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$
											$+S_2S_{1,j_2}{S_2}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1$
cont.								$+\frac{2\hbar\overline{\lambda_5}{L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\sqrt{3}M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^1$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1}{\sqrt{3}\epsilon}$
									$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1}{\sqrt{3}}$
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2}{\sqrt{3}\epsilon}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1}{\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2}{\sqrt{3}}$	$-\hbar\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_2}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_2}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$
								$+\frac{2\hbar\overline{\lambda_5}{L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\sqrt{3}\epsilon M_F}$	$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^1}{\epsilon}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_2}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$
											$+S_2S_{1,j_2}{S_2}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1$
cont.								$+\frac{2\hbar\overline{\lambda_5}{L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\sqrt{3}M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^1$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1}{\sqrt{3}\epsilon}$
									$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1}{\sqrt{3}}$
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2}{\sqrt{3}\epsilon}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1}{\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{\lambda_{HS_1}^{(2)}}^2}{\sqrt{3}}$	$-\hbar\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{\lambda_{12}^{(1)}}^1$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$S_1\,H$	$S_1\,S_2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_2 S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2 S_{1,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
								$+\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}\epsilon M_F}$	$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\epsilon}$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_2 S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
										$+S_2 S_{1,j_2}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}$
cont.								$+\frac{2\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_2 S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
									$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$	
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_2 S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_2 S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}}{\epsilon}$
										$-\hbar\overline{y_{2,i_1,r_1}}S_2 S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger}{3\epsilon}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\epsilon M_F}$	$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\epsilon}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$
										$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5}\overline{L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{3\epsilon}$
									$+H_{a_2}S_{1,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}\epsilon}$	$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
										$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}$
cont.										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$
										$-S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}$
										$+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^1\lambda_{12}^{(2)}$
cont.										$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
cont.										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$
										$+S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}$
cont.										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	$S_1\, H$	$S_1\, S_2$
											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$-S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}$
cont.											$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
cont.											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3$
cont.											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$
cont.											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$		$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger}{3\epsilon}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$		$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\epsilon M_F}$		$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1}{\epsilon}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger$
cont.								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{M_F}$		$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
										$+H_{a_2}S_{1,j_2}H_{a_1}{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2\lambda_{HS_1}^{(2)}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
											$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
											$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1$
cont.											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}}$
											$-S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2$
cont.											$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H ℓ	S_1 H	S_1 S_2
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
cont.										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$+{S_{1,j_4}S_{2,j_3}S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2$
cont.										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$-{S_{1,j_4}S_{2,j_3}S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2$
										$+{S_{1,j_3}S_{2,j_4}S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2$
cont.										$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
										$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
cont.										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
										$+{S_{1,j_3}S_{2,j_4}S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3$
cont.										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger}{3\epsilon}$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_4,i_2}}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger$
								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{\epsilon M_F}$	$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\epsilon}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger$
cont.								$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{\lambda_5 L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}{H_{a_1}}^\dagger{H_{a_2}}^\dagger}{M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$	$+\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
									$+H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$	$-\frac{4\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{3\epsilon}$
									$-\frac{2\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}\epsilon}$	$-\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}}{\sqrt{3}}$	$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$+\frac{4}{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	H	ℓ	S_1	S_2
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1$
cont.											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(1)}}^1}{\sqrt{3}}$
											$-S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2$
cont.											$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
cont.											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$+S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2$
cont.											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$-S_{1,j_4}S_{2,j_3}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2$
cont.											$+\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$+\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
cont.											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
											$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3$
											$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$
											$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
cont.											$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger {S_{2,j_2}}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$H\,\ell$	$S_1\,H$	$S_1\,S_2$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^{T}\,C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\epsilon M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}}{\sqrt{3}\epsilon}$	$-\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_1}{}^{\dagger}}{3\epsilon}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^{T}\,C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$	$-\frac{2h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}}{\sqrt{3}}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_1}{}^{\dagger}$
								$-\frac{2\sqrt{\frac{2}{3}}h\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^{T}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^{\dagger}H_{a_2}{}^{\dagger}}{\epsilon M_F}$	$+\frac{2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{1}}{\epsilon}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_1}{}^{\dagger}$
cont.								$-\frac{2\sqrt{\frac{2}{3}}h\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^{T}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^{\dagger}H_{a_2}{}^{\dagger}}{M_F}$	$+2i\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{1}$ $+H_{a_2}S_{1,j_2}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{2}\lambda^{(2)}_{HS_1}$ $-\frac{2h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{2}\lambda^{(2)}_{HS_1}}{\sqrt{3}\epsilon}$	$+\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$ $-\frac{4h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}}{3\epsilon}$
cont.								$-\frac{h\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{2}\lambda^{(2)}_{HS_1}}{\sqrt{3}}$	$-\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$ $+\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_2}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$ $+\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$ $+\frac{4}{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{1}\lambda^{(1)}_{12}$
cont.										$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{1}\lambda^{(1)}_{12}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{1}\lambda^{(1)}_{12}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{1}\lambda^{(1)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{1}\lambda^{(1)}_{12}}{\sqrt{3}}$ $-S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}$ $+\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$
cont.										$+\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$ $+\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$
cont.										$-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $+S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{2}\lambda^{(2)}_{12}$
cont.										$-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$ $-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $-\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $-S_{1,j_4}S_{2,j_3}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}$ $+S_{1,j_3}S_{2,j_4}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}$ $+\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$
cont.										$+\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $-\frac{2h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$
cont.										$-\frac{h\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}}$ $+\frac{2h\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^{2}\lambda^{(2)}_{12}}{\sqrt{3}\epsilon}$

Model α d_{S_1} Y_{S_1} d_{S_2} Y_{S_2} d_F Y_F H ℓ	S_1 H	S_1 S_2
		$+\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
cont.		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}\epsilon}$
		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$
		$+S_{1,j_3}S_{2,j_4}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3$
cont.		$-\frac{2\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
		$-\frac{\hbar\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$
cont.		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}\epsilon}$
		$-\frac{\hbar\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$

Field combinations 4 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1 F	S_1 ℓ	S_2 H
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2$ $+2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1}{\epsilon}$ $-2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2$ $+2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1}{\epsilon}$ $-2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_1^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left(S_1{}^\dagger\right)^2$ $+2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1y_{1,i_1,r_1}S_1{}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1^2y_{1,i_1,r_1}\left(S_1{}^\dagger\right)^2\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1}{\epsilon}$ $-2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}$	$\frac{3\hbar\overline{L_{a_1,i_1}}C\,P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}C\,P_R\,\overline{L_{a_2,i_2}}{}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}g_2^2}{8M_F}$ $+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_1}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_1}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2}{8M_F}$ $+\frac{3\hbar\overline{L_{a_1,i_1}}C\,P_R\,\overline{L_{a_2,i_2}}{}^T g_1^2S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}C\,P_R\,\overline{L_{a_2,i_2}}{}^T g_1^2S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{8M_F}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $+H_{a_2}S_{2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$

cont.

cont.

cont.

cont.

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,F$	$S_1\,\ell$	$S_2\,H$
									$+\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{2M_F}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{\lambda^{(2)}_{HS_2}}^2}{\epsilon}$
cont.									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{2\epsilon M_F}$	$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{\lambda^{(2)}_{HS_2}}^2$
									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{4M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_2}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{4M_F}$	
									$-\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{8M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}}{2M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{2\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{4M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{1,i_3,r_2}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{2M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{8M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}g_1^2\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{8\epsilon M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}g_1^2\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{8M_F}$	
cont.									$+\frac{\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger}{2M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}\lambda^{(1)}_{S_1}}{2\epsilon M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{I}_{y_1,a_2,j_2}\lambda^{(1)}_{S_1}}{2M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\lambda^{(1)}_{S_1}}{2\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_1F^*_{a_2,j_2}}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\lambda^{(1)}_{S_1}}{2M_F}$	
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+C_{{S_1}^\dagger S_1}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}{S_1}_{,j_1}{y_1}_{,i_1,r_1}{S_1}_{,j_1}{}^\dagger$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\epsilon}$
								$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}{S_1}_{,j_1}{S_1}_{,j_2}{y_1}_{,i_1,r_2}{y_1}_{,i_2,r_1}{S_1}_{,j_1}{}^\dagger{S_1}_{,j_2}{}^\dagger$	$-2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger$
								$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}{S_1}_{,j_3}{S_1}_{,j_4}{y_1}_{,i_1,r_1}{S_1}_{,j_1}{}^\dagger{S_1}_{,j_2}{}^\dagger\mathcal{I}_{{S_1},j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_1}$	$+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_2}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{\lambda^{(1)}_{HS_2}}^1}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{\lambda^{(1)}_{HS_2}}^1$
										$+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{\lambda^{(2)}_{HS_2}}^2}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{\lambda^{(2)}_{HS_2}}^2$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+\sqrt{3}C_{{S_1}^\dagger S_1}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}{S_1}_{,j_1}{y_1}_{,i_1,r_1}{S_1}_{,j_1}{}^\dagger$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
								$-\frac{5}{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}{S_1}_{,j_1}{S_1}_{,j_2}{y_1}_{,i_1,r_2}{y_1}_{,i_2,r_1}{S_1}_{,j_1}{}^\dagger{S_1}_{,j_2}{}^\dagger$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$
								$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}{S_1}_{,j_3}{S_1}_{,j_4}{y_1}_{,i_1,r_1}{S_1}_{,j_1}{}^\dagger{S_1}_{,j_2}{}^\dagger\mathcal{I}_{{S_1},j_1,j_3,j_2,j_4}{}^1\lambda^{(1)}_{S_1}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}\epsilon}$
										$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$
cont.										$+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_2}$
										$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_2}}{\epsilon}$
										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_2}$
cont.										$+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}$
										$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}}{\epsilon}$
cont.										$-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1 F	S_1 ℓ	S_2 H
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\sqrt{3}C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}_1$	$+\frac{3\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}g_2^2}{8M_F}$ $+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger g_2^2}{8M_F}$ $+\frac{3\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}g_1^2S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}g_1^2S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}}{8M_F}$ $+\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}}{2M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}}{2\epsilon M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}}{4M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_2}}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_2}}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{2,i_2,r_1}}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}}{\epsilon M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}}{2M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{1,i_3,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{2\epsilon M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{\epsilon M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{2M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{8M_F}$ $+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}g_1^2\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}g_1^2\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{8M_F}$ $+\frac{\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{2M_F}$ $-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}{\lambda_{S_1}^{(1)}}_1}{2\epsilon M_F}$ $-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}{\lambda_{S_1}^{(1)}}_1}{2M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger{\lambda_{S_1}^{(1)}}_1}{2\epsilon M_F}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathrm{Bar}(\mathrm{cg2})_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger{\lambda_{S_1}^{(1)}}_1}{2M_F}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}_1}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}_1}{\sqrt{3}}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\sqrt{3}C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{\lambda_{S_1}^{(1)}}_1$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}_1}{\sqrt{3}\epsilon}$ $-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}_1}{\sqrt{3}}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\epsilon}$ $-\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2y_{1,i_1,r_2}y_{1,i_2,r_1}\left({S_{1,j_1}}^\dagger\right)^2$ $+\frac{2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_2}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_2}^\dagger$ $+H_{a_2}S_2{H_{a_1}}^\dagger{S_2}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda_{HS_2}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1 F	S_1 ℓ	S_2 H
cont.									$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{S_1}}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{S_1}}{\sqrt{3}}$	$-\frac{2\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}}{\epsilon}$ $-\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S^2_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_1^\dagger S_1} \hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{S_1}}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{S_1}}{\sqrt{3}}$	$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_2{}^\dagger}{\epsilon}$ $-2\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_2{}^\dagger$ $+H_{a_2} S_2 H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}}{\epsilon}$ $-\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S^2_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_1^\dagger S_1} \hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{S_1}}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{S_1}}{\sqrt{3}}$	$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_2{}^\dagger}{\epsilon}$ $-2\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_2 y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_2{}^\dagger$ $+H_{a_2} S_2 H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}}{\epsilon}$ $-\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_2 y_{2,i_1,r_1} H_{a_1}{}^\dagger S_2{}^\dagger \mathcal{I}_{H2,a_1,a_2}{}^1 \lambda^{(1)}_{HS_2}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{2}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S^2_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_1^\dagger S_1} \hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{S_1}}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{S_1}}{\sqrt{3}}$	$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2i\sqrt{\frac{2}{3}} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$ $+2i\sqrt{\frac{2}{3}} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2} S_{2,j_2} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}}{\sqrt{3}}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S^2_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_1^\dagger S_1} \hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{S_1}}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{S_1}}{\sqrt{3}}$	$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2i\sqrt{\frac{2}{3}} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$ $+2i\sqrt{\frac{2}{3}} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2} S_{2,j_2} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}}{\sqrt{3}}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$+\frac{2}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S^2_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} \left(S_{1,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_1^\dagger S_1} \hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2}} S_{1,j_1} S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{S_1}}{\sqrt{3}}$ $-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{1,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger \mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{S_1}}{\sqrt{3}}$	$-\frac{2\hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_1} S_{2,j_1} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $+\frac{2i\sqrt{\frac{2}{3}} \hbar \overline{y_{2,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$ $+2i\sqrt{\frac{2}{3}} \hbar \overline{y_{2,i_1,r_1}} \overline{Y_{e,i_2,i_3}} H_{a_2} S_{2,j_2} y_{2,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$ $+H_{a_2} S_{2,j_2} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}$ $-\frac{2\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar \overline{y_{2,i_1,r_1}} H_{a_2} S_{2,j_2} y_{2,i_1,r_1} H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger \mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_2}}{\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1 F	S_1 ℓ	S_2 H
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}^2 y_{1,i_1,r_2}y_{1,i_2,r_1}\left({S_1}_{,j_1}^\dagger\right)^2$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
								$+\frac{2C_{S_1^\dagger S_1}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger}{\sqrt{3}}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$
								$-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger$		$+\frac{2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1}{\epsilon}$
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^1\lambda_{S_1}^{(1)}}{\sqrt{3}}$		$+2i\sqrt{\frac{2}{3}}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}{}^1$
cont.										$+H_{a_2}S_{2,j_2}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{1,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{1,j_2}}^\dagger\mathcal{I}_{S_1,j_1,j_3,j_2,j_4}{}^2\lambda_{S_1}^{(2)}}{\sqrt{3}}$		$-\frac{2\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$

Field combinations 5 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2\,F$	$S_2\,\ell$	$F\,B_{\mu\nu}$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$		$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2 y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{9}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
cont.									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$	
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$		$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2 y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
cont.									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2 y_{2,i_1,r_2}y_{2,i_2,r_1}\left(S_{2,j_1}{}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
cont.									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$	
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1		$+C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$+C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$	$+\frac{3\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}g_2^2}{8M_F}$ $+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger g_2^2}{8M_F}$	
cont.									$+\frac{3\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2 S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2 S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{8M_F}$ $+\frac{\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{2M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{4\epsilon M_F}$ $-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{8M_F}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{8M_F}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{2\epsilon M_F}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{4M_F}$	
cont.									$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_1^2 y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_1^2 y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2\,F$	$S_2\,\ell$	$F\,B_{\mu\nu}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$		$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+\sqrt{3}C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$		$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\sqrt{3}C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$	$+\frac{3\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}g_2^2}{8M_F}$ $+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger g_2^2}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger g_2^2}{8M_F}$
cont.									$+\frac{3\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{8\epsilon M_F}$ $+\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{8M_F}$ $+\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}$
cont.									$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_3}y_{1,i_3,r_1}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{4\epsilon M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_1}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_3}y_{2,i_3,r_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{4\epsilon M_F}$
cont.									$-\frac{3\sqrt{3}\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_3}y_{2,i_3,r_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{2M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{4M_F}$
cont.									$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}g_1^2y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8\epsilon M_F}$ $+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}g_1^2y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8M_F}$ $+\frac{\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{2M_F}$
cont.									$-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{2\epsilon M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4\epsilon M_F}$
cont.									$-\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8M_F}$ $-\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$ $-\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8M_F}$
cont.									$-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}\lambda_{S_2}^{(1)}}{2\epsilon M_F}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}\lambda_{S_2}^{(1)}}{2M_F}$ $-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{S_2}^{(1)}}{2\epsilon M_F}$
cont.									$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{S_2}^{(1)}}{2M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2\, F$	$S_2\, \ell$	$F\, B_{\mu\nu}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	ϵ_{ijk}	$+\sqrt{3}C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger$ $-\frac{5}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}$	$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	ϵ_{ij}	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_2}^\dagger\right)^2$ $+2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left({S_2}^\dagger\right)^2\lambda_{S_2}^{(1)}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	ϵ_{ij}	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_2}^\dagger\right)^2$ $+2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left({S_2}^\dagger\right)^2\lambda_{S_2}^{(1)}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	ϵ_{ij}	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_2^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_2}^\dagger\right)^2$ $+2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2y_{2,i_1,r_1}{S_2}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2^2y_{2,i_1,r_1}\left({S_2}^\dagger\right)^2\lambda_{S_2}^{(1)}$	$-\frac{9}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	ϵ_{ijk}	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_{2,j_1}}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$	$-\frac{9}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
cont.										
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	ϵ_{ijk}	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_{2,j_1}}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
cont.										
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	ϵ_{ijk}	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_{2,j_1}}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$
cont.										
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	ϵ_{ijk}	$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_2}^2y_{2,i_1,r_2}y_{2,i_2,r_1}\left({S_{2,j_1}}^\dagger\right)^2$ $+\frac{2C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_1}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{2,j_1}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger$	$-\frac{9}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)B_{\mu_1\mu_2}^2$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_2\,F$	$S_2\,\ell$	$F\,B_{\mu\nu}$
cont.								$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^1\lambda_{S_2}^{(1)}}{\sqrt{3}}$		
								$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{2,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{2,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{S_2,j_1,j_3,j_2,j_4}{}^2\lambda_{S_2}^{(2)}}{\sqrt{3}}$		

Field combinations 6 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,W_{\mu\nu}$	$H\,\ell\,e$	$S_1\,S_2\,H$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger$
									$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{\epsilon}$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_1\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\epsilon}$
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\sqrt{3}}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger$
									$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{\epsilon}$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_1\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\epsilon}$
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\sqrt{3}}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger$
									$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{\epsilon}$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_1}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_1\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\epsilon}$
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,W_{\mu\nu}$	$H\,\ell\,e$	$S_1\,S_2\,H$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\sqrt{3}}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1		$+\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$+\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}H_{a_1}{}^\dagger}{\epsilon}$ $-2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}H_{a_1}{}^\dagger$ $+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger{S_{1,j_1}}^\dagger$ $+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$ $-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_2}\mathcal{C}_{LS_1F^*_{a_2,j_1}}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
cont.										$-2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_2}\mathcal{C}_{LS_1F^*_{a_2,j_1}}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}H_{a_1}{}^\dagger\lambda^{(1)}_{HS_2}}{\epsilon}$ $+\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}H_{a_1}{}^\dagger\lambda^{(1)}_{HS_2}$
cont.										$+\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_1F^*_{a_2,j_2}}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(1)}_{HS_2}}{\epsilon}$ $+\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}H_{a_2}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_1F^*_{a_2,j_2}}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(1)}_{HS_2}$
cont.										$+\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}H_{a_1}{}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$ $+\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}H_{a_1}{}^\dagger\lambda^{(2)}_{HS_2}}{2\sqrt{3}}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}H_{a_1}{}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2F^*_{a_2,j_2}}H_{a_1}{}^\dagger\lambda^{(2)}_{HS_2}}{2\sqrt{3}}$
cont.										$+\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$ $+\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{HS_2}}{2\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	$W_{\mu\nu}$	$H\,\ell\,e$	$S_1\,S_2\,H$
cont.											$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_2}\mathcal{C}_{LS_1}{F^*}_{a_2,j_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}\epsilon}$
											$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_2}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_2}\mathcal{C}_{LS_1}{F^*}_{a_2,j_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{HS_2}}{2\sqrt{3}}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1			$+\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
										$+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
										$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{1}{4}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
											$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.											$-\frac{1}{4}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$		$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.											$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$		$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}{H_{a_1}}^\dagger}{\epsilon}$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-2\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}{H_{a_1}}^\dagger$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{S_2^*}_{a_1,a_2,j_1,j_2}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
											$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.											$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,W_{\mu\nu}$	$H\,\ell\,e$	$S_1\,S_2\,H$
										$-\frac{2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}\mathrm{cg4}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger}{\epsilon}$
cont.										$-2\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}\mathrm{cg4}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_1,j_2}{H_{a_1}}^\dagger\lambda^{(1)}_{HS_2}}{\epsilon}$
										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_1,j_2}{H_{a_1}}^\dagger\lambda^{(1)}_{HS_2}$
cont.										$-\frac{3\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}{H_{a_1}}^\dagger\lambda^{(1)}_{HS_2}}{\epsilon}$
										$-\frac{3}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}{H_{a_1}}^\dagger\lambda^{(1)}_{HS_2}$
										$-\frac{\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(1)}_{HS_2}}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(1)}_{HS_2}$
										$-\frac{3\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg4}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(1)}_{HS_2}}{\epsilon}$
										$-\frac{3}{2}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg4}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(1)}_{HS_2}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}{H_{a_1}}^\dagger\lambda^{(2)}_{HS_2}}{\epsilon}$
										$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}{H_{a_1}}^\dagger\lambda^{(2)}_{HS_2}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg4}_{a_1,j_1,a_2,j_2}H_{a_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{HS_2}}{\epsilon}$
										$+\frac{1}{2}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\mathrm{cg4}_{a_1,j_1,a_2,j_2}H_{a_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda^{(2)}_{HS_2}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\epsilon}$
cont.										$-\frac{1}{4}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+\overline{\lambda_5}S_2\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,W_{\mu\nu}$	$H\,\ell\,e$	$S_1\,S_2\,H$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+\overline{\lambda_5}S_2\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\sqrt{3}}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+H_{a_1}H_{a_2}S_{1,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger}{\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5{S_2}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+\overline{\lambda_5}S_2\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{3}{8}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\sqrt{3}}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\epsilon}$
cont.										$-\frac{1}{2}\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\,\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1{S_2}^*_{a_1,a_2,j_1,j_2}}{H_{a_1}}^\dagger{H_{a_2}}^\dagger{S_{1,j_1}}^\dagger}{2\sqrt{3}}$
									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger$
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger}{4\epsilon}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.									$+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\,\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\sqrt{3}}$
										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon}$
cont.										$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5{S_{2,j_1}}^\dagger}{2\sqrt{3}}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	$W_{\mu\nu}$	$H\,\ell\,e$	$S_1\,S_2\,H$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$		$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{2\sqrt{3}}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{2\sqrt{3}}$
cont.											
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$		$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{2\sqrt{3}}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{2\sqrt{3}}$
cont.											
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\left(W^{A_1}_{\mu_1\mu_2}\right)^2$		$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$ $+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$	$+\overline{\lambda_5}S_{2,j_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}\overline{\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{2\sqrt{3}}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}H_{a_1}{}^\dagger$	$+H_{a_1}H_{a_2}S_{1,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{2\sqrt{3}}$
cont.										$+\frac{\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger}{4\epsilon}$ $+\frac{3}{8}\sqrt{3}\hbar\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}H_{a_1}{}^\dagger$	$-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}\epsilon}$ $-\frac{\hbar\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{2\sqrt{3}}$
cont.											

Field combinations 7 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ S_2\ F$	$S_1\ S_2\ \ell$	$S_1\ F\ H$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_1}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}{S_1}^\dagger S_{2,j_1}{\mathcal{I}_{12,j_1,j_2}}^1\lambda_{12}^{(1)}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}{S_1}^\dagger S_{2,j_1}{\mathcal{I}_{12,j_1,j_2}}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{6\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{\epsilon M_F}$ $-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{M_F}$ $+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{M_F}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_1}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_1}^\dagger{\mathcal{I}_{H1,a_1,a_2}}^1\lambda_{H S_1}^{(1)}$
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{8M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{\epsilon M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{2M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{8M_F}$ $-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{8M_F}$ $-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{\sqrt{3}\epsilon M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{2\sqrt{3}M_F}$	
cont.									$-\frac{6\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_1^2S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{\epsilon M_F}$ $-\frac{2\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_1^2S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{M_F}$ $+\frac{\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{\epsilon M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{2M_F}$ $-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{8M_F}$ $-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{2\sqrt{3}M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{2,i_1,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{4\epsilon M_F}$ $-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$ $-\frac{h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2^*a_1,a_2,j_1}{S_{2,j_1}}^\dagger\lambda_{12}^{(1)}}{M_F}$	
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_1S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_1}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_1S_{2,j_2}y_{1,i_1,r_1}{S_1}^\dagger S_{2,j_1}{\mathcal{I}_{12,j_1,j_2}}^1\lambda_{12}^{(1)}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_1S_{2,j_2}y_{2,i_1,r_1}{S_1}^\dagger S_{2,j_1}{\mathcal{I}_{12,j_1,j_2}}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{M_F}$ $-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{4\epsilon M_F}$ $-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}{S_1}^\dagger}{8M_F}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_1y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_1}^\dagger$ $-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_1y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_1}^\dagger{\mathcal{I}_{H1,a_1,a_2}}^1\lambda_{H S_1}^{(1)}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{2M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{2\sqrt{3}M_F}$	
cont.									$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{2M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{2\sqrt{3}M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger \lambda^{(1)}_{12}}{M_F}$	
cont.									$-\frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger \lambda^{(1)}_{12}}{M_F}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}y_{2,i_2,r_2}} S_1 S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} {S_1}^\dagger {S_2}_{,j_1}{}^\dagger}{\sqrt{3}}$	$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{M_F}$	
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}} H_{a_1} S_1 y_{1,i_2,r_1} Y_{e,i_1,i_3} {H_{a_1}}^\dagger {S_1}^\dagger$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_1 y_{1,i_1,r_1} {H_{a_1}}^\dagger {S_1}^\dagger \mathcal{I}_{H1,a_1,a_2} {}^1\lambda^{(1)}_{H S_1}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_1 S_{2,j_2} y_{2,i_1,r_1} {S_1}^\dagger {S_2}_{,j_1}{}^\dagger \mathcal{I}_{12,j_1,j_2} {}^1\lambda^{(1)}_{12}}{\sqrt{3}}$		
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{2M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1} S_1{}^\dagger}{2\sqrt{3}M_F}$	
cont.									$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{2M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{2\sqrt{3}M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_1}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_1}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_1 y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1} {S_2}_{,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+\frac{3h\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger$	$-2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{Y_1,a_1,j_2}\mathcal{C}_{LS_1F^*_{a_2,j_1}}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$-2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$+\frac{h\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4M_F}$	$-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_1}^{(1)}$
								$-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$	$-\frac{9h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	$-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$
								$-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4M_F}$	
cont.								$+\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}$	$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{M_F}$	
								$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
								$+\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{8M_F}$	
cont.								$-\sqrt{3}h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda_{12}^{(2)}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{2\epsilon M_F}$	
								$-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4M_F}$	
								$-h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{2\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{C}_{LS_1F^*_{a_1,j_1}} \mathcal{C}_{LS_2F^*_{a_2,j_2}} S_{1,j_1}{}^\dagger}{4M_F}$	
									$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$	
cont.									$+\frac{3h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{9h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4M_F}$	
									$+\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{2\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{2\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{4M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}}{8M_F}$	
									$+\frac{3h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}}{4\epsilon M_F}$	
cont.									$+\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}}{4M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(1)}}{M_F}$	
cont.									$+\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
cont.									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$-C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}$	$+\frac{3h\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}g_2^2}{4\epsilon M_F}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_S\mathcal{C}_{LS_2F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}}{\sqrt{3}}$
								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}\mathrm{Bar(cg1)}_{j_3,j_4}$	$+\frac{h\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}g_2^2}{4M_F}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e_{i_2},i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1F^*_{a_2,j_1}}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}$
								$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar(cg1)}_{j_2,j_3}S_{1,j_1}{}^{\dagger}$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}g_2^2}{4\epsilon M_F}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\gamma_5}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_2}H_{a_1}{}^{\dagger}H_{a_2}{}^{\dagger}S_{1,j_1}{}^{\dagger}S_{1,j_2}{}^{\dagger}}{\sqrt{3}}$
								$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_1}\mathrm{Bar(cg1)}_{j_2,j_3}S_{2,j_1}{}^{\dagger}$	$+\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}g_2^2}{4M_F}$	
cont.									$-C_{S_2^\dagger S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$+\frac{3h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}g_2^2}{4\epsilon M_F}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{1}\lambda_{HS_1}^{(1)}$
								$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^{\dagger}g_2^2}{4M_F}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{1}\lambda_{HS_1}^{(1)}$
								$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$+\frac{3h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}g_2^2}{4\epsilon M_F}$	
cont.									$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^{T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}g_2^2}{4M_F}$	
								$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg1}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_3}{}^{\dagger}$	$+\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{4\epsilon M_F}$	
								$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\mathrm{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}$	$+\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{4M_F}$	
cont.									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg1}_{j_1,j_3}\mathrm{cg1}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{1,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}S_{2,j_4}{}^{\dagger}$	$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{M_F}$	
								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^{1}\lambda_{12}^{(1)}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{4\epsilon M_F}$	
								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^{1}\lambda_{12}^{(1)}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{8M_F}$	
cont.									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_2,j_3}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{2\epsilon M_F}$	
								$+\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\lambda_{12}^{(2)}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^{\dagger}}{4M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	$S_1\ S_2\ \ell$	$S_1\ F\ H$
											$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
cont.											$+\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{8M_F}$
											$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{\epsilon M_F}$
											$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger S_{2,j_3}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
cont.											$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
											$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$-\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{8M_F}$
											$+\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar}(\mathrm{cg1})_{j_2,j_3}S_{2,j_1}{}^\dagger\lambda_{S_2}^{(1)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.											$+\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg1}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger S_{2,j_3}{}^\dagger\lambda_{S_2}^{(1)}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{8M_F}$
												$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2\epsilon M_F}$
												$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.												$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
												$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
cont.												$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$
												$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
cont.												$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
												$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
												$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{\epsilon M_F}$
												$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$
cont.												$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
												$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$
												$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
cont.												$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
												$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$
												$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
cont.												$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
												$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
												$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
cont.												$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
												$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
												$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
cont.												$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$
												$-\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
											$+2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$+\frac{3\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$+\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
								$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1\lambda_{12}^{(1)}$	$-\frac{9\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$

$S_1\ F\ H$	
$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{\epsilon,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{\epsilon,i_1,i_3}\mathcal{I}_{y_1,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger$	
$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_1}^{(1)}$	
$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_1}^{(2)}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
								$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}\lambda_{12}^{(1)}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4M_F}$	
cont.								$+\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{M_F}$	
								$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
								$+\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$	
cont.								$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{2\epsilon M_F}$	
								$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.								$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2\lambda_{12}^{(2)}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{2\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$	
cont.									$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{9\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$+\frac{\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{3\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1F^*_{a_1,j_1}}\mathcal{C}_{LS_2F^*_{a_2,j_2}}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{C}_{LS_1F} {}^*_{a_1,j_1} \mathcal{C}_{LS_2F} {}^*_{a_2,j_2} S_{1,j_1}{}^\dagger \lambda^{(2)}_{12}}{\sqrt{3} M_F}$	
cont.									$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda^{(2)}_{12}}{\sqrt{3} \epsilon M_F}$	
									$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda^{(2)}_{12}}{\sqrt{3} M_F}$	
cont.									$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda^{(2)}_{12}}{\sqrt{3} \epsilon M_F}$	
									$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda^{(2)}_{12}}{\sqrt{3} M_F}$	
								$+\frac{2}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$-\frac{10}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$
								$-\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda^{(1)}_{12}$	$+\frac{9\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	$-\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1 \lambda^{(1)}_{HS_1}$
								$-\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^1 \lambda^{(1)}_{12}$	$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	
cont.								$+3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_2} S_{2,j_1} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda^{(2)}_{12}$	$-\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{2 M_F}$	$-\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{HS_1}$
								$-3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_1} S_{2,j_2} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda^{(2)}_{12}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
								$+3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_2} S_{2,j_1} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda^{(2)}_{12}$	$+\frac{3\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{16 M_F}$	
cont.								$-3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_2} S_{2,j_2} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \lambda^{(2)}_{12}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4 \epsilon M_F}$	
								$-\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda^{(2)}_{12}$	$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	
								$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
cont.								$-\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^2 \lambda^{(2)}_{12}$	$+\frac{3\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{16 M_F}$	
									$+\frac{\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{16 M_F}$	
cont.									$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{16 M_F}$	
cont.									$+\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{9\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
cont.									$+\frac{9\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$-\frac{\sqrt{3}\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{2 M_F}$	
cont.									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{9\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{16 M_F}$	
									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4 \epsilon M_F}$	
cont.									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{9\hbar \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{16 M_F}$	
cont.									$+\frac{3\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$+\frac{9\hbar \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{16 M_F}$	
									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 \epsilon M_F}$	
cont.									$+\frac{9\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{16 M_F}$	
									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4 M_F}$	
									$+\frac{3\hbar \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8 M_F}$	
cont.									$-\frac{3\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8 \epsilon M_F}$	
									$-\frac{\hbar \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8 M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$+\frac{9h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{3h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{2M_F}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{16M_F}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{16M_F}$	
cont.									$+\frac{\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{9\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{3\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{9\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}L_{a_1,i_1}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{2M_F}$	
									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$+\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{9h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{9h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$+\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$+\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+\frac{h\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {S_2}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3}M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-C_{S_2^{\frac{1}{2}}S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_1,j_2}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}g_2^2}{8\epsilon M_F}$	$-2\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}\lambda_5\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}$
								$-\frac{5}{9}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{1,j_3}S_{2,j_2}S_{2,j_4}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar(cg3)}_{j_1,j_2}\mathrm{Bar(cg3)}_{j_3,j_4}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}g_2^2}{8M_F}$	$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}}{\sqrt{3}}$
								$+\frac{10\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{1,i_2,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}S_{1,j_1}{}^{\dagger}}{3\sqrt{3}}$	$+\frac{9\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}g_2^2}{8\epsilon M_F}$	$-2\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{\lambda_5}\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}H_{a_1}{}^{\dagger}H_{a_2}{}^{\dagger}S_{1,j_1}{}^{\dagger}S_{1,j_2}{}^{\dagger}$
								$+\frac{10\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_2}y_{2,i_2,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}S_{2,j_1}{}^{\dagger}}{3\sqrt{3}}$	$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}g_2^2}{8M_F}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{\dagger}}{\sqrt{3}}$
cont.								$-C_{S_2^{\frac{1}{2}}S_2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$-\frac{3\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}{}^{T}S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}s_{2}{}^{*}_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^{\dagger}g_2^2}{8\epsilon M_F}$	$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{1}\lambda_{HS_1}^{(1)}$
								$+\frac{2}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}{}^{T}S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}s_{2}{}^{*}_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^{\dagger}g_2^2}{8M_F}$	$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^{\dagger}S_{1,j_1}{}^{\dagger}\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^{2}\lambda_{HS_1}^{(2)}$
								$-\frac{10}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$+\frac{9\sqrt{3}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}{}^{T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}g_2^2}{8\epsilon M_F}$	
cont.								$+\frac{8}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$+\frac{3\sqrt{3}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}{}^{T}S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^{\dagger}g_2^2}{8M_F}$	
								$-\frac{10}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
								$+\frac{10\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg3}_{j_2,j_3}S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{1,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}}{3\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8M_F}$	
cont.								$+\frac{10\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}\mathrm{cg3}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}}{3\sqrt{3}}$	$-\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{2M_F}$	
								$-\frac{5}{9}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}}\mathrm{cg3}_{j_1,j_3}\mathrm{cg3}_{j_2,j_4}y_{2,i_1,r_2}y_{2,i_2,r_1}S_{1,j_1}{}^{\dagger}S_{1,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}S_{2,j_4}{}^{\dagger}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_3}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
								$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^{1}\lambda_{12}^{(1)}$	$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{16M_F}$	
cont.								$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^{1}\lambda_{12}^{(1)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{4\epsilon M_F}$	
								$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{1,j_2}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}S_{1,j_1}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8M_F}$	
								$+3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
cont.								$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\lambda_{12}^{(2)}$	$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{16M_F}$	
								$+3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_3}y_{2,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{4M_F}$	
								$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_3}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8M_F}$	
cont.								$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_2,j_3}S_{1,j_1}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{1,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}\lambda_{12}^{(2)}}{\sqrt{3}}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_3}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^{\dagger}}{8M_F}$	
								$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^{2}\lambda_{12}^{(2)}$	$-\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}y_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
								$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}\mathcal{I}_{12,j_2,j_3,j_1,j_4}{}^{2}\lambda_{12}^{(2)}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}y_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8M_F}$	
cont.								$+\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}S_{2,j_3}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_2,j_3}S_{2,j_1}{}^{\dagger}\lambda_{S_2}^{(1)}$	$-\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{2M_F}$	
								$+\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_3}S_{2,j_2}y_{2,i_1,r_1}S_{1,j_1}{}^{\dagger}S_{2,j_2}{}^{\dagger}S_{2,j_3}{}^{\dagger}\lambda_{S_2}^{(1)}$		
cont.									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
									$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{16M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{4\epsilon M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
									$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{16M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
									$+\frac{9\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{16M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{8\epsilon M_F}$	
cont.									$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{16M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^{T}\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^{\dagger}}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$+\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$			
cont.													$-\frac{3h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
													$-\frac{h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8M_F}$			
													$-\frac{\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{2M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
cont.													$+\frac{3\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{16M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{16M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{2\epsilon M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{4M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
													$+\frac{3\sqrt{3}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{16M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{3\sqrt{3}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger}{16M_F}$			
													$-\frac{3\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
cont.													$-\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$-\frac{\sqrt{3}L_{a_1,i_1}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{2M_F}$			
													$+\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$			
cont.													$+\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$			
													$+\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
cont.													$+\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$			
cont.													$+\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$			
													$+\frac{9h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$			
													$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
cont.													$+\frac{\sqrt{3}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$			
													$+\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$			
cont.													$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$			
													$+\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$			
													$+\frac{h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$			
cont.													$+\frac{h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_1,j_2}}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
								$+\frac{2}{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8\epsilon M_F}$	$-\frac{4\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{10}{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1}{\sqrt{3}}$
								$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\mathcal{I}_{12,j_2,j_3,j_1,j_4}}^1\lambda_{12}^{(1)}$	$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8\epsilon M_F}$	$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1\lambda_{HS_1}^{(1)}$
								$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\mathcal{I}_{12,j_2,j_3,j_1,j_4}}^1\lambda_{12}^{(1)}$	$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	
cont.								$+3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$	$-\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{2M_F}$	$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger{\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^2\lambda_{HS_1}^{(2)}$
								$-3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	
								$+3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_2}S_{2,j_1}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$	$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{16M_F}$	
cont.								$-3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_1}S_{2,j_2}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
								$-\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger{\mathcal{I}_{12,j_2,j_3,j_1,j_4}}^2\lambda_{12}^{(2)}$	$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	
cont.									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{16M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{16M_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{16M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}^\dagger}{8M_F}$	
									$+\frac{9\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$	
cont.									$+\frac{9\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$	
									$-\frac{\sqrt{3}\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{2M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
									$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
									$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
									$+\frac{9\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{9\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{16M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{4\epsilon M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}^\dagger}{8M_F}$	
cont.									$-\frac{3\hbar\overline{L_{a_1,i_1}}\,C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{8\epsilon M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{8M_F}$	
									$+\frac{9\hbar\overline{L_{a_1,i_1}}\,C P_R\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{3\hbar\overline{L_{a_1,i_1}}\,C P_R\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{8M_F}$	
									$-\frac{\overline{L_{a_1,i_1}}\,C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{2M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{4\epsilon M_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{8M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}}\,C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}^\dagger}{8\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{16M_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4eM_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8eM_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{16M_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8eM_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{9\sqrt{3}\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{3\sqrt{3}\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{9\sqrt{3}\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
									$+\frac{3\sqrt{3}\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\sqrt{3}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{2M_F}$	
									$+\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4eM_F}$	
									$+\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_2,i_2}}^T C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8eM_F}$	
									$+\frac{9\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8eM_F}$	
cont.									$+\frac{9\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4eM_F}$	
									$+\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$+\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8eM_F}$	
									$+\frac{9\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8\epsilon M_F}$	
cont.									$+\frac{9\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{16M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$+\frac{\sqrt{3}\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{3}\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+\frac{\hbar\overline{y_{1,i_1,r_1}}\overline{y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
									$+\frac{\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}\epsilon M_F}$	
cont.									$+\frac{\hbar\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$		$-\frac{4\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_2S_{1,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_2{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	
									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_2S_{1,j_2}y_{1,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	
									$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_2S_{1,j_2}y_{2,i_1,r_1}S_2{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{12,j_1,j_2}{}^1\lambda_{12}^{(1)}$	
									$+\frac{\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{M_F}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{\sqrt{3}\epsilon M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$
									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{2\sqrt{3}M_F}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H_{S_1}}^{(2)}}{\sqrt{3}}$
									$-\frac{\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4eM_F}$	
cont.									$-\frac{3\sqrt{3}\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}C\,P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4eM_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1 S_2 F	S_1 S_2 ℓ	S_1 F H
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{2M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
cont.									$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{2\sqrt{3}M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{2M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger \lambda^{(1)}_{12}}{M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger \lambda^{(1)}_{12}}{M_F}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} {S_2}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$	$+\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{M_F}$	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_1}} S_{1,j_1} y_{1,i_2,r_1} {Y_{e,i_1,i_3}}^* {H_{a_1}}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_2 S_{1,j_2} y_{1,i_1,r_1} {S_2}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{12,j_1,j_2}{}^1 \lambda^{(1)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{\sqrt{3}\epsilon M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \overline{Y_{e,i_2,i_3}} H_{a_2}} S_{1,j_2} y_{1,i_2,r_1} {Y_{e,i_1,i_3}}^* {H_{a_1}}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{H_1,a_1,a_2,j_1,j_2}{}^1$
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_2 S_{1,j_2} y_{2,i_1,r_1} {S_2}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{12,j_1,j_2}{}^1 \lambda^{(1)}_{12}$	$-\frac{h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{2\sqrt{3}M_F}$	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} {H_{a_1}}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{H^\dagger_{S_1}}}{\sqrt{3}}$
cont.									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{\epsilon M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{2M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {s_2}^*_{a_1,a_2,j_1} {S_2}^\dagger}{8M_F}$	
cont.									$+\frac{\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{2\sqrt{3}M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{2M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger \lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{M_F}$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$-\frac{4\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}y_{2,i_2,r_2}} S_2 S_{1,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} {S_2}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{6h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{\epsilon M_F}$	$-\frac{2h\log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}} H_{a_1} S_{i,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} {H_{a_1}}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1}}_{,r_1} S_2 S_{1,j_2} y_{1,i_1,r_1} {S_2}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{12,j_1,j_2} {}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{2h\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}} H_{a_2} S_{i,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} {H_{a_1}}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{H_1,a_1,a_2,j_1,j_2} {}^1$
								$-2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_2 S_{1,j_2} y_{2,i_1,r_1} {S_2}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{12,j_1,j_2} {}^1\lambda_{12}^{(1)}$	$+\frac{\overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{M_F}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}} H_{a_2} S_{i,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} {H_{a_1}}^\dagger {S_{1,j_1}}^\dagger \mathcal{I}_{H_1,a_1,a_2,j_1,j_2} {}^2\lambda_{H S_1}^{(2)}}{\sqrt{3}}$
cont.									$-\frac{h\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{2\sqrt{3}M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{2M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_1}}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1 S_2 a_1,a_2,j_1}^* {S_2}^\dagger}{8M_F}$	
									$-\frac{6h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_1^2 S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_1^2 S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{M_F}$	
									$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{M_F}$	
									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T \sqrt{3}\epsilon M_F C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{2\sqrt{3}M_F}$	
cont.									$-\frac{\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\sqrt{3}h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} {S_{1,j_1}}^\dagger}{2M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1} S_2{}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$	
cont.									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1} S_2{}^\dagger \lambda^{(1)}_{12}}{M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger \lambda^{(1)}_{12}}{\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger \lambda^{(1)}_{12}}{M_F}$	
								$-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$	$-\frac{\sqrt{6}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{\epsilon M_F}$	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_1} S_{1,j_1} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_2} S_{2,j_1} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} Y_{e,i_2,i_3}} H_{a_2} S_{1,j_2} y_{1,i_2,r_1} Y_{e,i_1,i_3} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$
								$-\frac{4}{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2}} S_{1,j_1} S_{2,j_2} y_{1,i_2,r_1} y_{2,i_1,r_2} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{2\sqrt{6}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{\epsilon M_F}$	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger \mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2 \lambda^{(2)}_{H^\dagger_{S_1}}}{\sqrt{3}}$
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{12}}{\sqrt{3}}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$	
cont.								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1 \lambda^{(1)}_{12}}{\sqrt{3}}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$	
								$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.								$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
								$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.								$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_4} S_{2,j_3} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_4,j_3,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_4,j_3,j_2}{}^2 \lambda^{(2)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{1,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_4,j_3,j_2}{}^3 \lambda^{(3)}_{12}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{8M_F}$	
cont.								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} S_{1,j_3} S_{2,j_4} y_{2,i_1,r_1} S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger \mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^3 \lambda^{(3)}_{12}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{8M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	S_1	S_2	F	S_1	S_2	ℓ	S_1	F	H
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$			
cont.													$-\frac{\sqrt{6}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{\epsilon M_F}$			
													$-\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$			
													$-\frac{2\sqrt{6}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{\epsilon M_F}$			
cont.													$-\frac{2\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$			
													$-\frac{\sqrt{2}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$			
cont.													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$			
													$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$			
													$-\frac{\sqrt{2}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$			
cont.													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$			
cont.													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$			
													$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$			
													$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\epsilon M_F}$			
cont.													$-\frac{2h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$			
													$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\epsilon M_F}$			
													$-\frac{2h\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$			
cont.													$+\frac{\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
													$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$			
cont.													$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$			
													$+\frac{\sqrt{\frac{2}{3}}h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$			
cont.													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$			
													$-\frac{h\overline{y}_{1,i_1,r_1}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$			

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$+\frac{\sqrt{\frac{2}{9}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}{s_2}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{9}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}{s_2}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}{s_2}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}{s_2}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$	
cont.									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{7h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
									$-\frac{7h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{4}{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$	$-\frac{\sqrt{6}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{\epsilon M_F}$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}}{\sqrt{3}}$
								$+\frac{4}{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}\mathcal{I}_{H_1,a_1,a_2,j_1,j_2}{}^1$
								$-\frac{4}{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$+\frac{\sqrt{\frac{2}{3}}y_{1,i_1,r_1}y_{2,i_2,r_1}}{M_F}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger$	$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}{}^2\lambda_{H_{S_1}}^{(2)}}{\sqrt{3}}$
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.									$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
								$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$		
cont.									$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.									$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
									$-\frac{2\sqrt{3}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{2h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_2,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_3,r_2}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_1,r_1}y_{1,i_2,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3h\overline{y}_{1,i_3,r_1}y_{2,i_1,r_2}\overline{y}_{2,i_2,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_2}\overline{y}_{2,i_3,r_1}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y}_{1,i_1,r_1}y_{2,i_2,r_1}\overline{y}_{2,i_3,r_2}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$	
									$-\frac{\sqrt{6}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{M_F}$	
									$-\frac{\sqrt{2}h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
									$-\frac{2\sqrt{3}h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{2h\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$+\frac{\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$	
									$-\frac{h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{1,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y}_{2,i_3,r_1}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$+\frac{\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(2)}}{M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{7\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
									$-\frac{7\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_1}}^\dagger$	$-\frac{\sqrt{6}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}\overline{Y_{e,i_1,i_3}}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger}{\sqrt{3}}$
								$+\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$	$-\frac{\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger$	$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}\overline{Y_{e,i_1,i_3}}H_{a_1}{}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1$
								$-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger$	$+\frac{\sqrt{\frac{2}{3}}y_{i_1,r_1}y_{2,i_2,r_1}}{M_F}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}{H_{a_1}}^\dagger{S_{1,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^2\lambda_{H^\dagger S_1}^{(2)}}{\sqrt{3}}$
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{4\sqrt{2}M_F}$	
								$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3\sqrt{2}M_F}$	
cont.									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{4\sqrt{2}M_F}$	
								$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$		
cont.									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{4\sqrt{2}M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3\sqrt{2}M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{2\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{\epsilon M_F}$	
cont.								$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$	
								$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^3\lambda_{12}^{(3)}}{\sqrt{3}}$	$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{6M_F}$	
									$-\frac{\sqrt{6}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{\sqrt{\frac{2}{3}}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$	
									$-\frac{\sqrt{2}\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
									$-\frac{\sqrt{2}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_2,j_1}S_{2,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
									$-\frac{2\sqrt{3}\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{2\hbar\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$+\frac{\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{6M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{3h\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
cont.									$-\frac{h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3h\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
cont.									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$	
cont.									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
									$-\frac{h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\epsilon M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
									$-\frac{h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
cont.									$-\frac{7h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
									$-\frac{7h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
									$-\frac{7h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7h\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_1}{}^\dagger$	$-\frac{\sqrt{6}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{\epsilon M_F}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{1,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$
								$+\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$	$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H_{1,a_1,a_2,j_1,j_2}}{}^\dagger$
								$-\frac{4}{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}y_{2,i_1,r_2}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{2\sqrt{6}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{\epsilon M_F}$	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}S_{1,j_2}y_{1,i_2,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger\mathcal{I}_{H_{1,a_1,a_2,j_1,j_2}}{}^\dagger\lambda_{H^\dagger S_1}^{(2)}}{\sqrt{3}}$
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$		
cont.									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^1\lambda_{12}^{(1)}}{\sqrt{3}}$	$-\frac{2\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$+\frac{\sqrt{\frac{2}{3}}h\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
cont.									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$		
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
cont.									$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{\sqrt{2}h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\epsilon M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3\sqrt{2}M_F}$	
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_2,j_3,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$		
cont.									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	
								$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{1,i_1,r_1}S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{}^2\lambda_{12}^{(2)}}{\sqrt{3}}$	$-\frac{3h\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{4\sqrt{2}M_F}$	
									$-\frac{h\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2}\epsilon M_F}$	

Model α d_{S_1} Y_{S_1} d_{S_2} Y_{S_2} d_F Y_F $S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
	$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$	$-\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{4\sqrt{2}M_F}$
cont.	$+\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_4}S_{2,j_3}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
	$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_4,j_2}{\lambda_{12}^{(2)}}^2}{\sqrt{3}}$	$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{4\sqrt{2}M_F}$
	$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{1,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$	$-\frac{\sqrt{2}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{3\epsilon M_F}$
cont.	$-\frac{2\hbar\log\left(\frac{\mu^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}S_{1,j_3}S_{2,j_4}y_{2,i_1,r_1}{S_{1,j_1}}^\dagger{S_{2,j_2}}^\dagger\mathcal{I}_{12,j_1,j_3,j_2,j_4}{\lambda_{12}^{(3)}}^3}{\sqrt{3}}$	$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}{S_{1,j_1}}^\dagger}{3\sqrt{2}M_F}$
		$-\frac{2\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{\epsilon M_F}$
		$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{\sqrt{3}M_F}$
cont.		$-\frac{2\sqrt{3}\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{\epsilon M_F}$
		$-\frac{2\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{\sqrt{3}M_F}$
		$+\frac{\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{\sqrt{3}M_F}$
cont.		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{8M_F}$
		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{3\epsilon M_F}$
cont.		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{6M_F}$
		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{8M_F}$
cont.		$-\frac{\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{8M_F}$
		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{4\epsilon M_F}$
cont.		$-\frac{3\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{8M_F}$
		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{3\epsilon M_F}$
		$-\frac{\hbar\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}{L_{a_2,i_2}}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}{S_{1,j_1}}^\dagger}{6M_F}$
cont.		$-\frac{\sqrt{6}\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger}{\epsilon M_F}$
		$-\frac{\sqrt{\frac{2}{3}}\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger}{M_F}$
		$-\frac{2\sqrt{6}\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger}{\epsilon M_F}$
cont.		$-\frac{2\sqrt{\frac{2}{3}}\hbar\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}S_{2,j_1}{}^\dagger}{M_F}$
		$+\frac{\sqrt{\frac{2}{3}}\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{M_F}$
		$-\frac{\sqrt{2}\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{3\epsilon M_F}$
cont.		$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{3\sqrt{2}M_F}$
		$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{4\sqrt{2}M_F}$
cont.		$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{4\sqrt{2}M_F}$
		$-\frac{\sqrt{2}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{3\epsilon M_F}$
cont.		$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{3\sqrt{2}M_F}$
		$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{4\sqrt{2}M_F}$
cont.		$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{4\sqrt{2}M_F}$
		$-\frac{\sqrt{2}\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{3\epsilon M_F}$
cont.		$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{3\sqrt{2}M_F}$
		$-\frac{\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{4\sqrt{2}M_F}$
cont.		$-\frac{\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{2\sqrt{2}\epsilon M_F}$
		$-\frac{3\hbar\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2{}^*_{a_1,a_2,j_2,j_1}}{S_{2,j_1}}^\dagger}{4\sqrt{2}M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\,S_2\,F$	$S_1\,S_2\,\ell$	$S_1\,F\,H$
									$-\frac{2\sqrt{3}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{\epsilon M_F}$	
cont.									$-\frac{2\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{2\sqrt{3}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{\epsilon M_F}$	
									$-\frac{2\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
cont.									$+\frac{\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{6M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{6M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
									$-\frac{3\hbar\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$-\frac{\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4\epsilon M_F}$	
cont.									$-\frac{3\hbar\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
									$+\frac{\sqrt{\frac{2}{3}}\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
									$+\frac{\sqrt{\frac{2}{9}}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\epsilon M_F}$	
cont.									$+\frac{\sqrt{\frac{2}{9}}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}\epsilon M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$	
cont.									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$	
									$-\frac{\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
cont.									$-\frac{\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\epsilon M_F}$	
									$-\frac{3\sqrt{\frac{2}{11}}\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$	
cont.									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}\epsilon M_F}$	
									$-\frac{\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$	
									$-\frac{7\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7\hbar\overline{y_{1,i_1,r_1} y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	
									$-\frac{7\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}\epsilon M_F}$	
cont.									$-\frac{7\hbar\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$	

Field combinations 8 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ F\ \ell$	$S_2\ F\ H$	$S_2\ F\ \ell$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$	
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$	
T3-A	0	1	0	3	1	2	$\frac{1}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}}{\sqrt{3}}$	
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1		$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$	
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0		$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^\dagger C P_L L_{e_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}g_2^2}{8M_F}$ $+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger g_2^2}{8M_F}$ $+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^\dagger C P_L L_{e_1,i_1}g_1^2S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^\dagger C P_L L_{e_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^\dagger C P_L L_{e_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^\dagger C P_L L_{e_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}}{2M_F}$ $+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T g_1^2y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_{2,i_1,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{2M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_{2,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_{2,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^\dagger C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathcal{C}_{LS_2}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}\lambda_{S_2}^{(1)}}{2M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T y_{2,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{I}_{y_2,a_2,j_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{S_2}^{(1)}}{2M_F}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1		$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$	
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda_{HS_2}^{(2)}$	
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}{}^1}{\sqrt{3}}$ $-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{2,j_1}{}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda_{HS_2}^{(1)}$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^\dagger C P_L L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar}(\mathrm{cg1})_{a_1,j_1,a_2,j_2}g_2^2}{8M_F}$ $+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}{}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger g_2^2}{8M_F}$ $+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^\dagger C P_L L_{a_1,i_1}g_1^2S_{2,j_1}S_{2,j_2}\mathrm{Bar}(\mathrm{cg1})_{a_1,j_1,a_2,j_2}$ $8M_F$

cont.

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$S_1\ F\ \ell$	$S_2\ F\ H$	$S_2\ F\ \ell$
								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_2}y_{1,i_3,r_1}}{2M_F}$		$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_1}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{4M_F}$
cont.								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_2}y_{1,i_2,r_1}}{4M_F}$	$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}$	$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{1,i_3,r_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{4M_F}$
								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}}{M_F}$		$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}S_{2,j_2}y_{2,i_3,r_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}}{2M_F}$
								$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_1^2y_{2,i_1,r_1}y_{2,i_2,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{8M_F}$		
cont.								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{1,i_3,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathrm{Bar(cg2)}_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger$		$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{2,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{2M_F}$
								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{1,i_3,r_2}\mathrm{Bar(cg2)}_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{2M_F}$		$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$
								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathrm{Bar(cg2)}_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{M_F}$		$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg1}_{a_1,j_1,a_2,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}S_{2,j_1}{}^\dagger S_{2,j_2}{}^\dagger}{4M_F}$
cont.								$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_1}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}y_{2,i_3,r_2}\mathrm{Bar(cg2)}_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{4M_F}$		$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}S_{2,j_1}S_{2,j_2}\mathrm{Bar(cg1)}_{a_1,j_1,a_2,j_2}\lambda^{(1)}_{S_2}}{2M_F}$
								$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\bar{y}_1^2y_{1,i_3,r_2}\mathrm{Bar(cg2)}_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger}{8M_F}$		
								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T\mathrm{cg2}_{a_1,j_1,a_2,j_2}S_{1,j_1}S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_1}\lambda^{(1)}_{S_1}}{2M_F}$		
cont.								$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_1}}L_{a_2,i_2}{}^T\,C\,P_L\,L_{a_1,i_1}\mathrm{Bar(cg2)}_{a_1,j_1,a_2,j_2}S_{1,j_1}{}^\dagger S_{1,j_2}{}^\dagger\lambda^{(1)}_{S_1}}{2M_F}$		
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$	
									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}{Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_1,j_2}}^1}{\sqrt{3}}$	
									$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^1\lambda^{(1)}_{HS_2}$	
cont.									$-\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}$	
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$		$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_2}^\dagger$	
									$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_2y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_2}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda^{(1)}_{HS_2}$	
T3-D	0	3	0	1	1	2	$\frac{1}{2}$		$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_2}^\dagger$	
									$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_2y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_2}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda^{(1)}_{HS_2}$	
T3-D	2	3	1	1	2	2	$\frac{3}{2}$		$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_2y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_2}^\dagger$	
									$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_2y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_2}^\dagger\mathcal{I}_{H2,a_1,a_2}{}^1\lambda^{(1)}_{HS_2}$	
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$	
									$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}{Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^1$	
									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$	
									$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}{Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^1$	
									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$	
									$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}{Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^1$	
									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	
T3-E	2	3	1	3	2	2	$\frac{3}{2}$		$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_1}S_{2,j_1}y_{2,i_2,r_1}Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger}{\sqrt{3}}$	
									$+2i\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{2,j_2}y_{2,i_2,r_1}{Y_{e,i_1,i_3}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H1,a_1,a_2,j_2,j_1}}^1$	
									$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{2,j_2}y_{2,i_1,r_1}{H_{a_1}}^\dagger{S_{2,j_1}}^\dagger\mathcal{I}_{H2,a_1,a_2,j_1,j_2}{}^2\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	

Field combinations 9 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell$	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{\sqrt{3}}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{6\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}M_F}$ $-\frac{6\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T g_1^2S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{\sqrt{3}}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$ $+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C P_R\,\overline{L_{a_4,i_2}}{}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_1}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_1{}^\dagger}{\sqrt{3}}$ $-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$

cont.

									$-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}M_F}$ $-\frac{6\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T g_1^2S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^i S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C P_R\,\overline{L_{a_2,i_2}}{}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}a^*_{1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$

cont.

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1 S_2 F H	S_1 S_2 F ℓ
												$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$
cont.											$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_1y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger}{\sqrt{3}M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{\sqrt{3}M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_1{}^\dagger\lambda_{12}^{(1)}}{M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_1y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
								$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{9\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
									$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.											$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_1,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
												$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{9\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_1,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T g_2^2S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
cont.												$+\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
												$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_2,i_3}}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger g_2^2}{4M_F}$
								$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}C P_R\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger g_2^2}{4M_F}$
									$-\frac{1}{2}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}s^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}C P_R\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger g_2^2}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$
											$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger g_2^2}{4M_F}$
cont.											$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2 S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
											$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}\mathcal{I}_{y_2,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
											$+\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger\lambda^{(1)}_{HS_2}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
cont.											$+\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_1}\mathcal{C}_{LS_1}F^*_{a_2,j_2}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(1)}_{HS_2}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
											$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg1)}_{j_1,j_2}H_{a_1}{}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
											$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{y_1,a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}H_{a_1}{}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.											$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\mathrm{cg1}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{1,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
											$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_2}y_{2,i_1,r_1}\mathcal{I}_{y_2,a_1,j_2}\mathcal{C}_{LS_1}F^*_{a_2,j_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda^{(2)}_{HS_2}}{\sqrt{3}}$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda^{(1)}_{12}}{M_F}$
												$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda^{(1)}_{12}}{M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda^{(2)}_{12}}{\sqrt{3}M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger\lambda^{(2)}_{12}}{\sqrt{3}M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_2,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda^{(2)}_{12}}{\sqrt{3}M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger\lambda^{(2)}_{12}}{\sqrt{3}M_F}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C P_L L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}H_{a_2}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_2,r_1}y_{2,i_1,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2 S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
								$+\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{9\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2 S_{2,j_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
									$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.											$-\frac{1}{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{C}_{LS_1}F^*_{a_1,j_1}\mathcal{C}_{LS_2}F^*_{a_2,j_2}S_{1,j_1}{}^\dagger}{2M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C P_L L_{a_1,i_1}g_1^2 S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_1,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{9\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{4M_F}$
												$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{y_1,a_1,j_2}\mathcal{I}_{y_2,a_2,j_1}S_{2,j_1}{}^\dagger}{2M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1 S_2 F H	S_1 S_2 F ℓ
												$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
cont.												$+\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
												$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_2,r_1} y_{2,i_1,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{C}_{LS_1} F^*_{a_1,j_1} \mathcal{C}_{LS_2} F^*_{a_2,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
												$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{C}_{LS_1} F^*_{a_1,j_1} \mathcal{C}_{LS_2} F^*_{a_2,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
cont.												$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_2,r_1} y_{2,i_1,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
												$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{y_1,a_1,j_2} \mathcal{I}_{y_2,a_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{3} M_F}$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{3M_F}$	$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \lambda_5} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
								$-\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$	$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} \lambda_5} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{9\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
									$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} \lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} \lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{Ts,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$
												$+\frac{9\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{9\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{9\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} S_2^*{}_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$+\frac{9\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{9\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
cont.												$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
												$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}Y_{e,i_2,i_3}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_2,r_1}Y_{e,i_1,i_3}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}H_{a_1}{}^\dagger$	$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger g_2^2}{8M_F}$
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$	$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{9\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger g_2^2}{8M_F}$
									$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger g_2^2}{8M_F}$
									$-\frac{1}{2}\sqrt{3}\hbar\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{9\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger g_2^2}{8M_F}$
cont.											$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_1^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
									$-2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}Y_{e,i_2,i_3}}\mathrm{cg4}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_2,r_1}Y_{e,i_1,i_3}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg3)}_{j_1,j_2}H_{a_1}{}^\dagger\lambda_{HS_2}^{(1)}$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
cont.											$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_2}S_{2,j_1}y_{1,i_1,r_1}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}H_{a_1}{}^\dagger\lambda_{HS_2}^{(1)}$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
									$-\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg3}_{j_1,j_2}H_{a_1}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{HS_2}^{(1)}$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{2M_F}$
									$-3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg4}_{a_1,j_2,a_2,j_1}H_{a_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{HS_2}^{(1)}$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{8M_F}$
cont.											$+\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_2}S_{1,j_1}S_{2,j_2}y_{1,i_1,r_1}\mathrm{Bar(cg4)}_{a_2,j_1,a_1,j_2}H_{a_1}{}^\dagger\lambda_{HS_2}^{(2)}$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
									$+\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\mathrm{cg4}_{a_1,j_1,a_2,j_2}H_{a_2}y_{2,i_1,r_1}H_{a_1}{}^\dagger S_{1,j_1}{}^\dagger S_{2,j_2}{}^\dagger\lambda_{HS_2}^{(2)}$	$-\frac{3\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}g_1^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{8M_F}$
												$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
cont.												$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{2M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{2M_F}$
cont.												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{4M_F}$
cont.												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{8M_F}$
												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger}{2M_F}$
cont.												$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell$	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$
										$-\frac{3\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
cont.										$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
										$+\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1} H_{a_1} H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{3M_F}$	$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_2} y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
								$-\frac{2\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{3M_F}$	$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_2} y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$+\frac{9\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
									$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$
cont.									$-\frac{1}{2}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger$	$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} I_{T3,a_1,a_2,j_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$
										$+\frac{9\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_2^2 S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{9\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} g_1^2 S_{2,j_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{8M_F}$
										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2}\epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{9\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$+\frac{9\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
										$+\frac{9\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2}\epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell$	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$
cont.										$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{3\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{8M_F}$
cont.										$+\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$+\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
cont.										$+\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
										$+\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1,j_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{3}M_F}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_2{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{\sqrt{3}M_F}$
								$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_2{}^\dagger$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4M_F}$
									$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{M_F}$
cont.									$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4M_F}$
										$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$
										$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{\sqrt{3}M_F}$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_2{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{\sqrt{3}M_F}$
								$+\frac{2\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3}M_F}$	$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_1}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}\lambda_5 S_2{}^\dagger$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4M_F}$
									$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_2y_{1,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{M_F}$
cont.									$-\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_2y_{2,i_1,r_1}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1S_2}{}^*_{a_1,a_2,j_1}S_2{}^\dagger}{4M_F}$
										$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{\sqrt{3}M_F}$
										$-\frac{\sqrt{3}\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\tilde{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_2y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1}S_{1,j_1}{}^\dagger}{M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell$	$S_1\ S_2\ F\ H$	$S_1\ S_2\ F\ \ell$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger \lambda_{12}^{(1)}}{M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{\sqrt{3} M_F}$ $+\frac{2\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\, C\, P_R\, \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{\sqrt{3} M_F}$	$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 S_2{}^\dagger}{\sqrt{3}}$ $-\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_1} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} \lambda_5 S_2{}^\dagger$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \lambda_5} S_2 y_{1,i_1,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{6\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger}{M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger}{\sqrt{3} M_F}$ $-\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger}{4M_F}$
cont.									$-\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} \lambda_5} S_2 y_{2,i_1,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger$	$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger}{M_F}$ $-\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{1,i_1,r_2} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_2} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger}{4M_F}$
cont.										$-\frac{6\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} g_1^2 S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{\sqrt{3} M_F}$
cont.										$-\frac{\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1}^* S_2{}^\dagger \lambda_{12}^{(1)}}{M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_2 \mathcal{I}_{T3,a_1,a_2,j_1} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_4,i_2}{}^T C\, P_L\, L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}}\, C\, P_R\, \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} \lambda_5} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1,j_2}^* H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1} \lambda_5} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_1,j_2}^* H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{2\sqrt{6}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} g_1^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2} M_F}$
cont.									$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{2}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2} M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2} M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2} M_F}$ $-\frac{\sqrt{2}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{2\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} g_1^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{2\sqrt{3}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} g_1^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{3M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$ $-\frac{\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C\, P_L\, L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{\sqrt{6}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_2,j_1}^* S_{2,j_1}{}^\dagger}{M_F}$ $-\frac{2\sqrt{6}\hbar \log\left(\frac{\bar{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}}\, C\, P_R\, \overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1S_2} {a_1,a_2,j_2,j_1}^* S_{2,j_1}{}^\dagger}{M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,H\,\ell$	$S_1\,S_2\,F\,H$	$S_1\,S_2\,F\,\ell$
										$-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_3,r_1}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_2}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{2\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{2\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T g_1^2 S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_3,r_2}y_{2,i_1,r_1}y_{2,i_2,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_2}y_{2,i_3,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{3M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{1,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_2}y_{2,i_2,r_1}y_{2,i_3,r_2}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger}{4M_F}$
										$+\frac{\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$
cont.										$+\frac{\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}y_{2,i_2,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$-\frac{3\sqrt{\frac{2}{11}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$
										$-\frac{3\sqrt{\frac{2}{11}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_2,j_1}^*S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(2)}}{\sqrt{11}M_F}$
cont.										$-\frac{7\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$
										$-\frac{7\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{L_{a_1,i_1}}\,C\,P_R\,\overline{L_{a_2,i_2}}^T S_{1,j_1}y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_2}S_{2,j_1}{}^\dagger\lambda_{12}^{(3)}}{\sqrt{23}M_F}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_4,i_2}{}^T C\,P_L\,L_{a_2,i_1}H_{a_1}H_{a_3}\lambda_5\epsilon_{a_1,a_2}\epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{\lambda_5 L_{a_3,i_1}}\,C\,P_R\,\overline{L_{a_4,i_2}}^T y_{1,i_1,r_1}y_{2,i_2,r_1}\epsilon_{a_1,a_3}\epsilon_{a_2,a_4}H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\lambda_5}S_{2,j_2}y_{1,i_1,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\lambda_5}S_{2,j_2}y_{2,i_1,r_1}\mathcal{I}_{LLS_1}s_{2\,a_1,a_2,j_1,j_2}^*H_{a_1}{}^\dagger H_{a_2}{}^\dagger S_{1,j_1}{}^\dagger}{\sqrt{3}}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{1,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}y_2^2S_{2,j_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_1}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3M_F}$
cont.									$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}H_{a_1}H_{a_2}S_{1,j_2}y_{2,i_1,r_1}\mathcal{I}_{T3,a_1,a_2,j_2,j_1}\lambda_5 S_{2,j_1}{}^\dagger}{\sqrt{3}}$	$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{1,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\sqrt{2}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}y_{2,i_3,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_2}y_{2,i_3,r_2}\mathcal{I}_{T3,a_1,a_2,j_1,j_2}S_{1,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{2\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}y_2^2S_{2,j_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{M_F}$ $-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_2}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_1}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_3,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}y_{1,i_2,r_2}y_{2,i_3,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}y_{2,i_1,r_2}y_{2,i_2,r_1}}L_{a_2,i_2}{}^T C\,P_L\,L_{a_1,i_1}S_{2,j_1}y_{1,i_3,r_2}\epsilon_{a_1,a_2}S_{1,j_1}{}^\dagger}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,H\,\ell$	$S_1\,S_2\,F\,H$	$S_1\,S_2\,F\,\ell$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{4M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{3M_F}$
										$-\frac{\sqrt{6}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{M_F}$
										$-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{3M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger}{2\sqrt{2}M_F}$
										$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T y_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_3} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.										$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
cont.										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$
										$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(2)}}{M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$
cont.										$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$
										$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$
										$-\frac{7\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
cont.										$-\frac{7\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$ $-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} H_{a_1}{}^\dagger H_{a_2}{}^\dagger}{M_F}$	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{\lambda_5} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} \overline{\lambda_5} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$ $-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{\sqrt{6}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} y_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{M_F}$ $-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_1} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3M_F}$
cont.									$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger}{\sqrt{3}}$	$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$ $-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3M_F}$
cont.										$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} y_2^2 S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{M_F}$ $-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{4M_F}$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	F	H	ℓ	S_1	S_2	F	H	S_1	S_2	F	ℓ
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{3M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{4M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{4M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{4M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger}{3M_F}$			
															$-\frac{\sqrt{6}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{M_F}$			
															$-\frac{\sqrt{2}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{3M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$			
															$-\frac{\sqrt{2}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{3M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_1,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger}{2\sqrt{2}M_F}$			
															$-\frac{2\sqrt{3}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{3M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{4M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{4M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{3M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{4M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger}{4M_F}$			
cont.															$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
															$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger \lambda_{12}^{(1)}}{M_F}$			
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger \lambda_{12}^{(1)}}{\sqrt{3}M_F}$			
															$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(2)}}{M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
cont.															$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_2,j_1} {S_{2,j_1}}^\dagger \lambda_{12}^{(2)}}{M_F}$			
															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger \lambda_{12}^{(2)}}{\sqrt{11}M_F}$			
															$-\frac{7\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
cont.															$-\frac{7\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} {S_{2,j_1}}^\dagger \lambda_{12}^{(3)}}{\sqrt{23}M_F}$			
T3-E	2	3	1	3	2	2	$\frac{3}{2}$	$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_4,i_2}}^T C P_L L_{a_2,i_1} H_{a_1} H_{a_3} \lambda_5 \epsilon_{a_1,a_2} \epsilon_{a_3,a_4}}{M_F}$		$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} \overline{\lambda_5} S_{2,j_2} y_{1,i_1,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$		$-\frac{\sqrt{6}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{M_F}$	
								$-\frac{2\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{\lambda_5 L_{a_3,i_1}} C P_R \overline{L_{a_4,i_2}}^T y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_3} \epsilon_{a_2,a_4} {H_{a_1}}^\dagger {H_{a_2}}^\dagger}{M_F}$		$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} \overline{\lambda_5} S_{2,j_2} y_{2,i_1,r_1} \mathcal{I}_{LLS_1} {S_2}^*_{a_1,a_2,j_1,j_2} {H_{a_1}}^\dagger {H_{a_2}}^\dagger {S_{1,j_1}}^\dagger}{\sqrt{3}}$		$-\frac{2\sqrt{6}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{M_F}$	
										$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{1,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger}{\sqrt{3}}$		$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} {S_{1,j_1}}^\dagger \lambda_{12}^{(3)}}{2\sqrt{2}M_F}$	
cont.															$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{2,i_1,r_1}} H_{a_1} H_{a_2} S_{1,j_2} y_{2,i_1,r_1} \mathcal{I}_{T3,a_1,a_2,j_2,j_1} \lambda_5 {S_{2,j_1}}^\dagger}{\sqrt{3}}$		$-\frac{\sqrt{2}\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{3M_F}$	
																	$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$	
																	$-\frac{\hbar \log\left(\frac{\tilde{\mu}^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{1,i_2,r_2} y_{2,i_2,r_1}} {L_{a_2,i_2}}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{1,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} {S_{1,j_1}}^\dagger}{2\sqrt{2}M_F}$	

Model α d_{S_1} Y_{S_1} d_{S_2} Y_{S_2} d_F Y_F F H ℓ	S_1 S_2 F H	S_1 S_2 F ℓ
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{2\sqrt{2} M_F}$
cont.		$-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} y_{2,i_3,r_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger}{3M_F}$
		$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_2^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{M_F}$
		$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} g_1^2 S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{3M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{1,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{1,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} y_{2,i_1,r_2} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_2} y_{2,i_3,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{4M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1} y_{2,i_3,r_2}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger}{3M_F}$
cont.		$-\frac{\sqrt{6}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{M_F}$
		$-\frac{2\sqrt{6}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{M_F}$
		$-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{3M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{2\sqrt{2} M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{2\sqrt{2} M_F}$
		$-\frac{\sqrt{2}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{3M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_2} y_{1,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{2\sqrt{2} M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger}{2\sqrt{2} M_F}$
		$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T g_2^2 S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{M_F}$
cont.		$-\frac{2\sqrt{3}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T g_1^2 S_{1,j_1} y_{1,i_1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{1,i_3,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_3,r_2} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_3,r_2} y_{2,i_1,r_1} y_{2,i_2,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_2} y_{2,i_3,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{3M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{1,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{2,i_3,r_1} L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_2} y_{2,i_2,r_1} y_{2,i_3,r_2} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger}{4M_F}$
		$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
cont.		$+\frac{\sqrt{\frac{2}{3}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{i,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(1)}}{\sqrt{3} M_F}$
		$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_2} \mathcal{I}_{T3,a_1,a_2,j_1,j_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$
cont.		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
		$-\frac{3\sqrt{\frac{2}{11}}\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_2} y_{1,i_1,r_1} y_{2,i_2,r_1} \mathcal{I}_{LLS_1 S_2{}^*_{a_1,a_2,j_2,j_1}} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{M_F}$
		$-\frac{\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(2)}}{\sqrt{11} M_F}$
cont.		$-\frac{7\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{y_{1,i_1,r_1} y_{2,i_2,r_1}} L_{a_2,i_2}{}^T C P_L L_{a_1,i_1} S_{2,j_1} \epsilon_{a_1,a_2} S_{1,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$
		$-\frac{7\hbar \log\left(\frac{\mu^2}{M_F^2}\right) \overline{L_{a_1,i_1}} C P_R \overline{L_{a_2,i_2}}{}^T S_{1,j_1} y_{1,i_1,r_1} y_{2,i_2,r_1} \epsilon_{a_1,a_2} S_{2,j_1}{}^\dagger \lambda_{12}^{(3)}}{\sqrt{23} M_F}$

Field combinations 10 of 10

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell\ e$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	-4	1	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	-2	1	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-A	0	1	0	3	1	2	$\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	1	-1	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	1	0	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-B	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	1	1	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	-3	2	$-\frac{3}{2}$	2	$-\frac{1}{2}$	3	-1	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}}\overline{L_{a_1,i_2}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}}\overline{Y_{e,i_3,i_2}}\overline{e_{i_2}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\,H\,\ell\,e$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	-1	2	$-\frac{1}{2}$	2	$\frac{1}{2}$	3	0	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-C	1	2	$\frac{1}{2}$	2	$\frac{3}{2}$	3	1	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	-2	3	-1	1	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	0	3	0	1	1	2	$\frac{1}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-D	2	3	1	1	2	2	$\frac{3}{2}$	$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-E	-4	3	-2	3	-1	2	$-\frac{3}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-E	-2	3	-1	3	0	2	$-\frac{1}{2}$	$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.								$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}\overline{e_{i_2}}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$

Model	α	d_{S_1}	Y_{S_1}	d_{S_2}	Y_{S_2}	d_F	Y_F	$F\ H\ \ell\ e$	
									$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-E	0	3	0	3	1	2	$\frac{1}{2}$		$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
									$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.									$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$
									$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{1,i_2,r_1}Y_{e,i_3,i_1}$
T3-E	2	3	1	3	2	2	$\frac{3}{2}$		$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_3,r_1}\overline{L_{a_1,i_2}}}\,P_R\,e_{i_1}H_{a_1}y_{2,i_2,r_1}Y_{e,i_3,i_1}$
									$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{1,i_1,r_1}\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}}\,P_L\,L_{a_1,i_1}y_{1,i_3,r_1}{H_{a_1}}^\dagger$
cont.									$+\frac{1}{4}\sqrt{3}\hbar\log\left(\frac{\bar{\mu}^2}{M_F^2}\right)\overline{y_{2,i_1,r_1}\overline{Y_{e,i_3,i_2}}\,\overline{e_{i_2}}}\,P_L\,L_{a_1,i_1}y_{2,i_3,r_1}{H_{a_1}}^\dagger$